

SESSION

5



Average

Learning Objectives

1. Average
2. Weighted Average

SESSION 05 – Statistics I

by PANKAJ BURHANPURKAR



STATISTICS – *Extraction of meaningful information* is studied in a branch of mathematics called **Statistics**.

The word 'statistics' appears to have been derived from the Latin word '**status**' meaning '**a (political) state**'. In its origin, statistics was simply the collection of data on different aspects of the life of people, **useful to the State**.

Over the period of time, however, its scope broadened and statistics began to concern itself not only with the collection and presentation of data but also with the interpretation and drawing of inferences from the data.

Statistics deals with collection, organisation, analysis and interpretation of data.

Data – Raw data, Facts or figures, which are numerical or otherwise, collected with a definite purpose are called data.

Data is the plural form of the Latin word datum.

Ex. Attendance, Cricket Score Board, Picnic Supper

Organisation of Data

Tally Marks, Table, Arrangement of Data in ascending or descending order

Grouping of Data

Frequency of Data Points |
 Ungrouped frequency distribution
 table or simply a Frequency
 Distribution Table | Grouped
 Frequency Distribution | Lower
 Class Limit (LCL) | Upper Class
 Limit (UCL) | Class Interval or Class
 Width or Class Size (UCL – LCL) |
 Boundary (Subtract 0.5 from LCL
 and Add 0.5 in UCL) | Class Mark
 $\{(UL + LL)/2\}$

Marks Obtained	Tally Marks	Frequency
12	I	1
15	II	2
17	III	3
19	IIII	5
21	III	3
23	IIII	4
25	II	2

H	I	J	K	L	M	N
		Class Interval	Class Boundary	Class Mark	Frequency	
		70-74	69.5-74.5	72	8	
		75-79	74.5-79.5	77	8	
		80-84	79.5-84.5	82	5	
		85-89	84.5-89.5	87	11	
		90-94	89.5-94.5	92	9	
		95-99	94.5-99.5	97	5	
		100-104	99.5-104.5	102	4	
					50	

Arithmetic Mean (or Average)



Rule 1 - Average lies between minimum and MAXIMUM.

Find the average of

1. $\{10, 20, 35, 40, 80\}$

2. $\{30, 30, 30, 30, 30\}$

3. $\{20, 30, 30, 32, 38\}$

$$x > 4$$

Quantity A
Average (x , 4)

Quantity B
4

Quantity A

Average (11, 19, x , 40)

Quantity B

Average (10, 20, 40, x)

Rule 2 - Average is the process of EQUALLING the data. Suppose, the average of a group is X , then we can say that every member of that group can be equal to X .

The weekly production of four wheelers in 7 weeks is as follows:
233, 239, 236, 231, 230, 224, 227
Calculate average weekly production of four wheelers

Rule 3 –

1. “Each observation $+ X$ ” , then average will be increased by X
2. “Each observation $- X$ ” , then average will be decreased by X
3. “Each observation $\times X$ ” , then average will be multiplied by X
4. “Each observation $\div X$ ” , then average will be divided by X

List A = { 10,20,30,40,50 } . If 10 is added to each observation, then find the change in average .

Present average age of a group of boys is 25 years. Find the average age of the group after 7 years?

If Average amount with some boys is 20. If there is decrease of 100 in their whole amount, then new average becomes 17.5 then find number of boy?

Rule 4 –

When a number is inserted in the group

List A = {10, 20, 45, 75, 90, 48}

Average = 48

If average is inserted then there is no change in the average.

If more than average is inserted then average increases.

If less than average is inserted then average decreases.

When a number is removed from the group

List A = {10, 20, 45, 75, 90, 48}

Average = 48

If average is removed then there is no change in the average.

If more than average is removed then average decreases.

If less than average is removed then average increases.

Average height of 80 boys is 104 cm. 2 boys with height 98 cm and 110 cm leaves the group then, find the new average?

The average height of a group of 12 is 82 cm. When two men with height 72 cm and 94 cm are replaced by men with height 106 cm and 48 cm then, find new average height?

The Average amount with x boys is Rs. 36. When a boy with Rs. 72 joins the group, then the average amount increases by Rs. 2. Find x ?

The average of the first 4 numbers out of 5 numbers is 62 and the average of the last 4 numbers is 67. If the ratio of first and last numbers is 7:9; then find the last number?

Ten years ago the average age of a couple was 27 years. Today the present average age of the couple and their child is 27 years. Find the present age of the child?

Rule 5 –

Arithmetic Sequence - An arithmetic sequence or arithmetic progression (AP) is a sequence whose terms go up or down by constant steps i.e., there is a common difference.

If data is in AP then Median = Mean.

If data is in AP Mean = (First Term + Last Term)/2

Examples –

Find Average of 5, 7, 9, 11, 13

Find Average of 88, 78, 68, 58, 48

Find Average of 100, 150, 200, 250, 300

Average of 25 consecutive natural numbers is 25. Find the smallest number?

Find average of all the multiples of 7 between 100 and 350?



Thanks!

SESSION 05 – Statistics I

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Weighted Average

$$C_{\text{avg}} = \frac{C_1 w_1 + C_2 w_2 + \dots}{w_1 + w_2 + \dots}$$

C_{avg} is average of combined group/mixture

$C_1, C_2, \dots$ are average of individual groups.

$w_1, w_2, \dots$ are ratio of group members or quantities.

$w_1, w_2, \dots$ are ratio of group members or quantities.

Class A - 20 Boys- Average height 40 units | 30 Girls average height 50 units

Class B - 20 Boys- Average height 50 units | 30 Girls average height 40 units

Class C - 20 Boys- Average height 40 units | 20 Girls average height 50 units

Class D - 20 Boys- Average height 40 units | 30 Girls average height 40 units

In what ratio, should Raj mix Rs. 40 per kg wheat with Rs. 60 per kg wheat so that the mixture's cost price becomes Rs. 55 per kg?

In what ratio, should Raj mix Rs. 40 per kg wheat with Rs. 60 per kg wheat so that Cost Price of mixture become if he sells wheat at 25% profit at Rs. 55 per kg.

Ratio of Milk and Water in Mixture A and B are 5 : 3 and 11 : 5 respectively. Find ratio of milk and water if mixture A and B are mixed equally.

A merchant has 1000 kg of sugar, part of which he sells at 8% profit and the rest at 18% profit. He gains 14% on the whole. The quantity sold at 18% profit is:

Answer Options: 400 kg, 560 kg, 600 kg, 640 kg

A person has Rs. 5000. He invests a part of it at 3% per annum and the remainder at 8% per annum simple interest. His total income in 3 years is Rs. 750. Find the sum invested at different rate of interests?

A person covers a distance of 100 kms in 10 hours, partly by walking at 7 km/hr and the rest by running at 12 km/hr. Find the distance covered in each part.

A milk vendor has 2 cans of milk. The first contains 25% water and the rest milk. The second contains 50% water. How much milk solution should he mix from each of the containers so as to get 12 lit solution such that the ratio of water to milk is 3:5?

(Mark all the correct options)

Answer Options: 4 lit, 5 lit, 6 lit, 8 lit, 7 lit

In a class average height of x boys is 80cm & average height of y girls is 64cm and average height of entire class is not more than 72cm then which of the following can be the values for (x,y) ? (Mark all the correct answers)

Answer Options:

(500,500), (25,51), (51,25), (1,100), (72,70)

In a class average weight of 35 boys is x kg and that of 54 girls is y kg and an average weight of entire class is z kg then

Quantity A

$$|x-z|$$

Quantity B

$$|y-z|$$



Thanks!

Class Practice Exercise 21

Statistics I

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. The average age of a group of 23 persons increases by 4 months, when a person of 20 years is replaced by a new person. The age of the new person is
 (1) 28 years (2) 27 years 8 months
 (3) 27 years (4) 26 years 6 months
2. In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs?
 (1) 6.25 (2) 6.5
 (3) 6.75 (4) 7
3. The average age of a group of 23 persons increases by 4 months, when a person of 20 years is replaced by a new person. The age of the new person is
 (1) 28 years (2) 27 years 8 months
 (3) 27 years (4) 26 years 6 months
4. The second of the three numbers is twice the first and third is 4 times the first. If the average of the three numbers is 7, then find the smallest of the three numbers.
 (1) 1 (2) 3
 (3) 7 (4) 6
5. The average age of a group of officers goes down to 56 years from 58 years, when an old member is replaced by a young member. What is the total number of members in that team if the age difference between those two members is 30 years?
 (1) 12 (2) 15
 (3) 20 (4) 18
6. A family consists of two grandparents, two parents and three grandchildren. The average age of the grandparents is 67 years, that of the parents is 35 years and that of the grandchildren is 6 years. What is the average age of the family?
 (1) $28\frac{4}{7}$ years (2) $31\frac{5}{7}$ years
 (3) $32\frac{1}{7}$ years (4) None of these
7. Average weight of 10 people is 50 kg. When one person is added to the group the average weight increases by 1 kg. So, what is the weight of that new person?
 (1) 60 kg (2) 59 kg
 (3) 61 kg (4) 50 kg
8. The average height of 13 people reduces by 2 cm if a person of height 184 cm is replaced by a new person. Find the height of the new person?
 (1) 154 cm (2) 159 cm
 (3) 197 cm (4) 158 cm
9. A grocer has a sale of Rs. 6435, Rs. 6927, Rs. 6855, Rs. 7230 and Rs. 6562 for 5 consecutive months. How much sale must he have in the sixth month so that he gets an average sale of Rs. 6500?
 (1) Rs. 4991 (2) Rs. 5991
 (3) Rs. 6001 (4) Rs. 6991
10. The average weight of A, B and C is 40 kg. If the average weight of A and B is 35 kg and that of B and C is 36 kg, then find the weight of B.
 (1) 22 kg (2) 23 kg
 (3) 28 kg (4) 30 kg
11. There are 24 students in a class whose average marks in a subject, the maximum marks of which is 100, is 89. If 3 students leave the class, then what is the maximum by which the average could go up?
 (1) 10.7 (2) 10.5
 (3) 11.2 (4) 11
12. The average of 20 numbers is zero. Of them, at the most, how many may be greater than zero?
 (1) 0 (2) 1
 (3) 10 (4) 19
13. Mean of 20 observations was 63 but later it was found that the observation 69 was misread as 96. Find the correct mean.
 (1) 62 (2) 61.65
 (3) 62.65 (4) 64.35
14. A shipping clerk has five boxes of different but unknown weights each weighing less than 100 kg. The clerk weights the boxes in pairs. The weights obtained are 110, 112, 113, 114, 115, 116, 117, 118, 120 and 121 kg. What is the weight of the heaviest box?
 (1) 60 kg (2) 65 kg
 (3) 64 kg (4) 62 kg
15. The average weight of 8 person's increases by 2.5 kg when a new person comes in place of one of them weighing 65 kg. What might be the weight of the new person?
 (1) 76 kg (2) 76.5 kg
 (3) 85 kg (4) Data inadequate
16. Out of 4 numbers, whose average is 60, the first one is one-fourth of the sum of the last three. The first number is
 (1) 15 (2) 45
 (3) 48 (4) 60
17. There are 40 boys in a class. One of them weighing 100 kg goes away. A new boy joins the class at the same time. Consequently, the average weight of the boys is increased by 0.25kg. Find the weight of the new boy.
 (1) 98.5 kg (2) 110 kg
 (3) 105 kg (4) 99 kg

18. The captain of a cricket team of 11 members is 26 years old and the wicket keeper is 3 years older. If the ages of these two are excluded, the average age of the remaining players is one year less than the average age of the whole team. What is the average age of the team?
 (1) 23 years (2) 24 years
 (3) 25 years (4) None of these
19. The average marks of two sections A and B of mathematics of class IX in the annual examination is 74. The average marks of section A is 77.5 and that of section B is 70. The ratio of the number of students of sections A and B is
 (1) 7 : 8 (2) 7 : 5
 (3) 8 : 7 (4) 8 : 5
20. The class teachers of three classes X, Y and Z give an algebra test to their respective students. The average scores in class X, Y and Z are 83, 76 and 85 respectively. The average score of all students in classes X and Y together is 79 and the average score of all students in classes Y and Z together is 81. What is the average for all the three classes?
 (1) 81 (2) 81.5
 (3) 82 (4) 84.5
21. The average monthly income of P and Q is Rs. 5050. The average monthly income of Q and R is Rs. 6250 and the average monthly income of P and R is Rs. 5200. The monthly income of P is –
 (1) 3500 (2) 4000
 (3) 4050 (4) 5000
22. In a family, the average age of a father and a mother is 35 years. The average age of the father, mother and their only son is 27 years. What is the age of the son?
 (1) 12 years (2) 11 years
 (3) 10.5 years (4) 10 years
23. There are 'n' students in a class. The average age of a student of this class is 'A' years. Two students left the class and 4 new students joined. Incidentally, the average age of a student in the class, did not change. The sum of the ages of the 6 students (who either left or joined the class) was 72 years. Find the value of A.
 (1) 9 (2) 12
 (3) 15 (4) 18
24. The average age of husband, wife and their child 3 years ago was 27 years and that of wife and the child 5 years ago was 20 years. The present age of the husband is –
 (1) 35 years (2) 40 years
 (3) 50 years (4) None of these
25. A is the average of a series of n consecutive odd numbers starting with x and B is the average of n consecutive odd numbers starting with x + 4. The value of (B – A)
 (1) is always 4. (2) depends on x
 (3) depends on n (4) is always a multiple of 4
26. The average rainfall for Monday, Tuesday and Wednesday is 4.3 cm. The average rainfall for Friday, Saturday and Sunday is 3.9 cm. If the average rainfall for the total week is 3.7 cm, then what is the rainfall recorded on Thursday?
 (1) 1.4 cm (2) 1.3 cm
 (3) 1.2 cm (4) 1.1 cm
27. A car owner buys petrol at Rs.7.50, Rs. 8 and Rs. 8.50 per litre for three successive years. What approximately is the average cost per litre of petrol if he spends Rs. 4000 each year?
 (1) Rs. 7.98 (2) Rs. 8
 (3) Rs. 8.50 (4) Rs. 9
28. The average age of a husband and wife who were married 5 years ago, was 25 years at that time. The average age of the family including the husband, the wife and two children who were born during the interval is 16 years now. How old are the children now? [All of them have different ages with integral values]
 (1) 1 year, 3 years (2) 4 years, 1 year
 (3) 2 years, 3 years (4) 2 years, 2 years
29. The average age of 5 members of a committee is the same as it was 3 years ago, because an old member has been replaced by a new member. The difference between the ages of old and new member is:
 (1) 15 years (2) 5 years
 (3) 3 years (4) 8 years
30. In Arun's opinion, his weight is greater than 65 kg but less than 72 kg. His brother does not agree with Arun and he thinks that Arun's weight is greater than 60 kg but less than 70 kg. His mother's view is that his weight cannot be greater than 68 kg. If all are correct in their estimation, what is the average of different probable weights of Arun?
 (1) 67 kg (2) 68 kg
 (3) 69 kg (4) Data inadequate
31. The average weight of a class of 100 students is 45 kg. The class consists of two sections, A and B, each with 50 students. The average weight W1 of Section A is 1 kg less than the average weight W2 of Section B. If Deepak, the heaviest student of section B, is moved to Section A, and Poonam, the lightest student of Section A, is moved to Section B, then the average weights of the two sections are interchanged. What is the weight of Poonam if it is known that movement of Deepak from Section B to A (without moving Poonam) makes the average weights of the two sections equal?
 (1) 70 kg (2) 15 kg
 (3) 20 kg (4) 25 kg
32. The average speed of a train for the first 5 hours was 63 km/hr, for next 3 hours was 58 km/hr and for next 2 hours was 72 km/hr. What was the average speed of the train for the entire journey?
 (1) 65 km/hr (2) 63.3 km/hr
 (3) 60 km/hr (4) 64.33 km/hr

33. The average weight of A, B and C is 45 kg. If the average weight of A and B be 40 kg and that of B and C be 43 kg, then the weight of B is –
 (1) 17 kg (2) 20 kg
 (3) 26 kg (4) 31 kg
34. Students of two different sections of a school appeared for a test. The average score of the students of the 1st section is 65 and the average score of the boys and the girls of the 1st section are 68 and 64 respectively. The average score of the students of the 2nd section is 71 and the average scores of the boys and the girls of the 2nd section are 80 and 65 respectively. If the number of girls in the 2nd section is half the number of boys in the 1st section, then the average score of the students of the two sections together is
 (1) 67.7 (2) 66.6
 (3) 69.2 (4) 66
35. Two similar vessels are filled with the mixtures of water and milk in the ratio of 3 : 4 and 5 : 3 respectively. If the mixtures are poured into a third vessel, the ratio of water and milk in the third vessel will be
 (1) 15 : 12 (2) 53 : 59
 (3) 20 : 9 (4) 59 : 53
36. In what ratio must a mixture of 30% alcohol strength be mixed with another mixture of 50% alcohol strength so as to get a mixture of 45% alcohol strength?
 (1) 1 : 2 (2) 1 : 3
 (3) 2 : 1 (4) 3 : 1
37. A shopkeeper mixed 10 kg of rice costing Rs.10 per kg with a certain quantity of another variety of rice costing Rs.15 per kg. He sold the mixture at the rate of Rs.14 per kg. If he gained 5%, the quantity of second variety of rice mixed was
 (1) 10 kg (2) 15 kg
 (3) 20 kg (4) 30 kg
38. A person covers 800 km partly at a speed of 50 kmph and partly at a speed of 150 kmph, in 10 hours over all. What is the distance covered at the speed of 150 kmph?
 (1) 240 km (2) 350 km
 (3) 560 km (4) 450 km
39. Ratios of milk and water in two solutions are 2:3 and 3:2 respectively. Content of these two solutions are mixed in the ratio 1:4. Find the ratio of milk and water in the resultant solution.
 (1) 12 : 13 (2) 14 : 11
 (3) 13 : 2 (4) 11 : 14
40. If certain quantity of water, which equals to 10% of the volume of the solution, is added to a solution of milk and water, and then 10% of the solution is removed, the ratio of milk to water becomes 2 : 3. Find the ratio of milk to water in the original solution.
 (1) 11 : 14 (2) 14 : 11
 (3) 3 : 8 (4) 2 : 7
41. An empty container is filled with pure alcohol. The alcohol is slowly allowed to run out and when the container is $\frac{1}{4}$ th empty, it is filled with water. Next when the container is half empty it is again filled with water. Finally, when it is $\frac{3}{4}$ th empty, it is again filled with water. Find the final percentage of alcohol in the container.
 (1) $8\frac{1}{2}\%$ (2) $11\frac{3}{4}\%$
 (3) $9\frac{3}{8}\%$ (4) $14\frac{3}{8}\%$
42. A cylindrical container $\frac{4}{7}$ th is full. From this, if x litres is removed, the container becomes only $\frac{1}{4}$ th full. If 35 L is now added to the container it will become half full. Find the value of x and the capacity of the container respectively.
 (1) 45 L and 210 L (2) 25 L and 140 L
 (3) 45 L and 140 L (4) 40 L and 210 L
43. Two alloys of iron have different percentages of iron in them. The first of them weighs 6 kg and the second one weighs 12 kg. One piece each, of equal weight, was cut off from both the alloys. The piece from the first was mixed with the leftover part of the second alloy and the piece from the second with the leftover part of the first alloy. As a result the percentage of iron became same in both the alloys. What was the weight of each piece cut off?
 (1) 4 kg (2) 3.5 kg
 (3) 3 kg (4) None of these
44. In a class there are 12 boys and 24 girls. Average weight of boys and girls is 50 kg and 40 kg respectively. Then what is the average weight of the class?
 (1) 42.33 kg (2) 43.33 kg
 (3) 44.33 kg (4) 45.33 kg
45. Average rainfall from Monday to Saturday is 5 inches and the average for the whole week is 8 inches. What is the rainfall on Sunday?
 (1) 23 inches (2) 25 inches
 (3) 27 inches (4) None of these
46. Average height of 5 people is 162cm. When one person is removed, then the average height of the remaining people becomes 161cm. What is the height of the person removed?
 (1) 161 cm (2) 162 cm
 (3) 163 cm (4) None of these
47. Ram lives along with his wife, son and daughter-in-law. The average age of Ram's family 3 years ago was 40 years. Three years later Ram dies due to illness at the age of 53 years and at the same time his daughter-in-law gave birth to Ram's grandson. If all the ages are always taken as integral value, now what is the average age of Ram's family?
 (1) 24.80 years (2) 26.45 years
 (3) 29.75 years (4) 31.90 years

48. The average of 'n' numbers is x. When 36 is subtracted from two of the numbers, the new average becomes (x - 8). The value of 'n' is
 (1) 6 (2) 8
 (3) 9 (4) 72
49. What is the Average of 23, 25, 29, 37, 40?
 (1) 29.5 (2) 30.2
 (3) 30.8 (4) 31.5
50. Sanjay purchased three dozen Banana at Rs. 15 per dozen, two dozen Banana at Rs. 22.50 per dozen and five dozen Banana at Rs. 24 per dozen. The average cost per dozen of the Banana that he purchased is
 (1) Rs. 20.00 (2) Rs. 20.75
 (3) Rs. 21.00 (4) Rs. 21.30
51. The average number of runs scored by Tendulkar in the last 10 matches is 42. In the last 15 matches Ganguly scored runs at an average rate of 57. The average of runs scored by both of these batsmen is
 (1) 49.5 (2) 50.4
 (3) 51.0 (4) 52.2
52. I bought 5 kgs of rice costing Rs. 32 per kg and another 7 kgs of rice costing Rs. 50 per kg. What is the average cost price per kg of rice I bought?
 (1) 41.00 (2) 41.75
 (3) 42.50 (4) 43.00
53. A trader bought 8 kgs of rice costing Rs. 15 per kg and 22 kgs of rice costing Rs. 25 per kg. What is his average cost price per kg?
 (1) Rs. 20.00 (2) Rs. 21.80
 (3) Rs. 22.33 (4) Rs. 23.00
56. Amit borrowed Rs. 6000 at rate of 4% and Rs. 4000 at rate of 6%. If the interest rates are simple interest rates, find the average rate Amit has to pay on the entire borrowed amount.
 (1) 4.6% (2) 4.7%
 (3) 4.8% (4) 4.9%
54. Nine liters of milk and water solution having 40% milk is mixed with 3 liters of milk and water solution having 70% milk. Find the percentage of milk in the mixture of the two solutions.
 (1) 45.0% (2) 46.8%
 (3) 47.5% (4) 50.0%
55. One-third of the goods are sold at a profit of 10%. At what profit percent should the rest of the goods be sold so as to earn an overall profit of 20%?
 (1) 22.5% (2) 24%
 (3) 25% (4) 27.5%
56. Part of goods worth Rs. 15,000 were sold at a profit of 6% and rest at a profit of 24%. If the overall profit on the entire amount is 21%, goods worth how much was sold at 6% profit?
 (1) Rs. 2000 (2) Rs. 2250
 (3) Rs. 2500 (4) Rs. 3000
57. Two solutions having ratio of milk and water in the ratio 2:3 and 1:5 are mixed in ratio of volumes of 3:5. Find the ratio of milk and water in the resultant mixture.
 (1) 59 : 181 (2) 60 : 180
 (3) 61 : 179 (4) 63:177
58. How many litres of solution having milk and water in ratio 8:1 should be mixed with 12 litres of solution having ratio of milk and water in the ratio 5:1 so that the resultant mixture has milk and water in the ratio 7:1.
 (1) 30 litres (2) 32 litres
 (3) 36 litres (4) 40 litres
59. How many liters of water needs to be added to 10 litres of a solution having milk and water in the ratio 3:2 to result in a solution having milk and water in the ratio 2:3?
 (1) 3 litres (2) 4 litres
 (3) 5 litres (4) 6 litres
60. A 20-liter mixture of milk and water contains milk and water in the ratio 3:2. 10 litres of the mixture is removed and replaced with pure milk and the operation is repeated once more. At the end of the two removals and replacement, what is the ratio of milk and water in the resultant mixture?
 (1) 7 : 3 (2) 8 : 2
 (3) 9 : 1 (4) 17 : 3
61. A mixture of 125 gallons of wine and water contains 20% water. How much water (in gallons) must be added to the mixture in order to increase the percentage of water to 25% of the new mixture?
 (1) 10 (2) 8.33
 (3) 8 (4) 6.66
62. A cistern contains 50 liters of water, 5 liters of water is taken out of it and replaced by wine. The process is repeated again. Find the proportion of wine and water in the resulting mixture.
 (1) 1:4 (2) 41:50
 (3) 19:81 (4) 81:19
63. A mixture of 20 liters of brandy and water contains 10% water. How much water should be added to it to increase the percentage of water to 25%?
 (1) 2 litres (2) 3 litres
 (3) 2.5 litres (4) 4 litres
64. Two solutions of 90% and 97% purity are mixed resulting in 21 litres of mixture of 94% purity. How much is the quantity of the first solution in the resulting mixture?
 (1) 15 litres (2) 12 litres
 (3) 9 litres (4) 6 litres
65. A gain percent of 20 is made by selling the mixture of two types of ghee at Rs. 480 per kg. If the type costing 610 per kg was mixed with 126 kg of the other, how many kilograms of the former was mixed?
 (1) 84 kg (2) 90 kg
 (3) 96 kg (4) 105 kg

66. A vendor mixes 'x' kg of rice cost 0.5\$ per kg with 20 kg of another rice costing 0.2\$ per kg. The mixture is sold at 0.36\$ per kg with a profit of 20%. Find x.
 (1) 8 kg (2) 12 kg
 (3) 10 kg (4) 15 kg
67. In what ratio must a grocer mix two varieties of tea-worth Rs. 60 a kg and Rs. 65 a kg so that by selling the mixture at Rs. 68.20 a kg he may gain 10%?
 (1) 3:2 (2) 3:4
 (3) 3:5 (4) 1:2
68. How many liters of 35% acid solution should be mixed with 15 liters of 42% acid solution so that the mixture has 40% acid.
 (1) 5 litres (2) 6 litres
 (3) 7.5 litres (4) 9 litres
69. A milk solution vendor mixes milk solution from 2 cans of milk solution to get total 33 liters. The first contains 62% water and the rest milk. The second contains 84% water. How much milk solution should he mix from each of the containers so as to get 33 liters of milk solution such that the ratio of water to milk in mixture is 7:3?
 (1) 18 L of first and 15 L of second
 (2) 21 L of first and 12 L of second
 (3) 16.5 L of first and 16.5 L of second
 (4) 2.4 L of first and 9 L of second
70. 15 liters of 10% spirit solution is mixed with x liters of 6% spirit solution. If the resulting solution has 9% concentration, then find x.
 (1) 12 lit (2) 18 lit
 (3) 21 lit (4) 5 lit
71. A milk vendor has 2 cans of milk. The first contains 25% water and the rest milk. The second contains 50% water. How much milk should he mix from each of the containers so as to get 12 liters of milk such that the ratio of water to milk is 3:5? Mark one or more correct options
 (1) 4 lit (2) 5 lit
 (3) 6 lit (4) 7 lit
72. A color dye is prepared by adding red and yellow colors in the ratio 2:5. If the quantity of red color in the dye is increased by 25%, determine the percentage of yellow color in the new dye formed.
 (1) 70% (2) 60%
 (3) 66.67% (4) 71.4%
73. In what ratio must water be mixed with milk to gain 12.5% on selling the mixture at cost price?
 (1) 1 : 5 (2) 1 : 10
 (3) 1 : 7 (4) 1 : 8
74. Two vessels of equal capacity are completely filled with mixture of water and alcohol. One contains water to alcohol in ratio 7:5 and other contains water to alcohol in ratio 7:8. If the contents of the two vessels are mixed. What is the ratio of water to alcohol in the mixture so formed?
 (1) 14 : 13 (2) 20 : 17
 (3) 22 : 19 (4) 21 : 19
75. A man invests \$5000 in two banks. Bank A gives 11% rate while bank B gives 15% rate. If he gets \$700 as interest in first year. Determine the amount invested in bank B.
 (1) \$1000 (2) \$1250
 (3) \$2000 (4) \$3750
76. 4 kg of a metal contains copper to zinc in ratio 1:4. A second metal of 5 kg contains zinc to copper in ratio 5:1. If these two metals are mixed, determine the ratio of copper to zinc in the mixture so formed
 (1) 11:49 (2) 49:221
 (3) 94:181 (4) 39:231
77. In 75 lit of a mixture ratio of milk to water is 4:1. How much water should be added to this mixture that the ratio of milk to water becomes 3:1?
 (1) 3 litres (2) 5 litres
 (3) 6 litres (4) 7.5 litres
78. A container contains 40 liters of milk. From this container 4 liters of milk taken out and replaced with water. This process was repeated further two times. How many liters of milk is now contained by the container?
 (1) 26.34 (2) 27.36
 (3) 28 (4) 29.16
79. A vessel is filled with liquid, 3 parts of which are water and 7 parts milk. How much part of the mixture must be drawn off and replaced with water so that the mixture will have 50% milk?
 (1) 1/5 (2) 3/10
 (3) 2/7 (4) 1/4
80. A container has water and acid mixed in ratio 11:5. When 8 liters of the mixture was drawn and replaced by water, the ratio of water to acid became 49:15. Determine the quantity of acid in the mixture at the beginning.
 (1) 12 litres (2) 8 litres
 (3) 10 litres (4) 15 litres
- DIRECTIONS for Q. 81 to 85:** Mark your answer as, option
 (1) If Quantity A is greater than B
 (2) If Quantity B is greater than A
 (3) If both quantities are equal
 (4) If A and B are not comparable with the information given.
81. A mixture contains nothing but water and acetone in the ratio of 1:2. After 200 ml of water is added the mixture, ratio of water to acetone is 2:3.
Quantity A: The original volume of mixture.
Quantity B: 1800 ml (c)
82. Steve has a 40-ounce mixture of apple juice and seltzer that contains 30% apple juice. If he pours 10 more ounces of apple juices into the mixture.
Quantity A: What % of mixture will be seltzer.
Quantity B: 50% (a)

83. A full glass of juice in a mixture of 20% grape juice and rest apple juice. The contents of the glass are poured into a pitcher that is 200% larger than the glass. The remainder of the pitcher is filled with 16 ounces of water.

Quantity A: Original volume of grape Juice in mixture

Quantity B: 1.6 ounce (c)

84. At a certain school, the 118 juniors have an average final exam score of 88 and the 100 seniors have an average final exam of 92.

Quantity A: The average final exam score of all the Juniors and Seniors combined.

Quantity B: 90 (b)

85. Mocktail 'X' has 55% alcohol, Mocktail 'Y' has 70% alcohol. Thor combines both mocktails in a single glass such that it contains 65% alcohol.

Quantity A: Weight of mocktail 'X'

Quantity B: Weight of mocktail 'Y'. (b)

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

86. Consider a class of 40 students whose average weight is 40 kgs. m new students join this class whose average weight is n kgs. If it is known that $m + n = 50$, what is the maximum possible average weight of the class now?

- (1) 40.18 kgs (2) 40.56 kgs
(3) 40.67 kgs (4) 40.49 kgs

87. The average score in an examination of 10 students of a class is 60. If the scores of the top five students are not considered, the average score of the remaining students falls by 5. The pass mark was 40 and the maximum mark was 100. It is also known that none of the students failed. (4) If each of the top five scorers had distinct integral scores, the maximum possible score of the topper is

- (1) 99 (2) 100
(3) 87 (4) 95

88. The ratio of a two-digit natural number to a number formed by reversing its digits is 4 : 7. Which of the following is the sum of all the numbers of all such pairs?

- (1) 99 (2) 198
(3) 330 (4) 132

89. In a class of 5 students, average weight of the 4 lightest students is 40 kgs, Average weight of the 4 heaviest students is 45 kgs. What is the difference between the maximum and minimum possible average weight overall?

- (1) 2.8 kgs (2) 3.2 kgs
(3) 3 kgs (4) 4 kgs

90. Janta Airline has a free luggage allowance for its passengers. If any passenger carries excess luggage, it is charged at a constant rate per kg. The total luggage charge paid by Ravind Jekriwal and Pranas Shubhan is Rs. 1100. If both Ravind and Pranas had carried luggage twice the weight than they actually did, their luggage charges would have been Rs. 2000 and Rs. 1000

respectively. What was the charge levied on Ravind's luggage?

- (1) Rs. 800 (2) Rs. 700
(3) Rs. 600 (4) Rs. 900

91. 1 unit of $x\%$ alcohol is mixed with 3 units of $y\%$ alcohol to give 60% alcohol. If $x > y$, how many integer values can x take?

- (1) 19 (2) 20
(3) 21 (4) 13

92. Class A has boys to girls in the ratio 2 : 3, Class B has girls to boys in the ratio 5 : 3. If the number of students in Class A is at least twice as many as the number of students in Class B, what is the minimum percentage of boys when both classes are considered together?

- (1) 33.33% (2) 40%
(3) 39.17% (4) 37.5%

93. School X has 3 classes, A, B and C. The average score of Class B is 16 more than the average score of class C. The strengths of Classes A, B and C are in the ratio 2:3:5, respectively. If the average score of the school is 2 less than the average score of Class A, Find the difference between average of class A and class C.

- (1) 8.5 (2) 7.5
(3) 6.5 (4) 5.5

94. Natural numbers 1 to 25 (both inclusive) are split into 5 groups of 5 numbers each. The medians of these 5 groups are A, B, C, D and E. If the average of these medians is m , what are the smallest and the largest values m can take?

- (1) 8, 18 (2) 9, 17
(3) 6, 16 (4) 5, 15

95. Consider 4 numbers a, b, c and d . Ram figures that the smallest average of some three of these four numbers is 30 and the largest average of some three of these 4 is 40. What is the range of values the average of all 4 numbers can take?

- (1) 12.5 (2) 25
(3) 37.5 (4) 50

96. 100 kgs of an alloy of tin and lead in the ratio 1:3 is mixed with x kgs of an alloy of tin and lead in the ratio 3:2. If the overall alloy should contain between 40% and 50% tin, what is the range of values x can take?

- (1) $100 \text{ kgs} \leq x \leq 200 \text{ kgs}$
(2) $80 \text{ kgs} \leq x \leq 240 \text{ kgs}$
(3) $110 \text{ kgs} \leq x \leq 220 \text{ kgs}$
(4) $75 \text{ kgs} \leq x \leq 250 \text{ kgs}$

97. Ten years ago, the ages of the members of a joint family of eight people added up to 231 years. Three years later, one member died at the age of 60 years and a child was born during the same year. After another three years, one more member died, again at 60, and a child was born during the same year. The current average age of this eight-member joint family is nearest to

- (1) 22 years (2) 24 years
(3) 26 years (4) 28 years

98. The average weight of 16 boys in a class is 50.25 kg and that of the remaining 8 boys is 45.15 kg. Find the average weights of all the boys in the class.
 (1) 47.55 kg (2) 48 kg
 (3) 48.55 kg (4) 49.25 kg
99. A, B, C and D are four friends, each of whom weighs less than 100 kg. From amongst them they form a group and find the total weight of that group. If the sum of the total weights of all possible distinct groups, each having the same number of members as in the first is 882 kg, then what is the average weight of the four friends?
 (1) 41.75 kg (2) 63.5 kg
 (3) 73.5 kg (4) Can't be determined
100. The average marks obtained by the candidates who passed and those who failed in an examination are 60 and 25 respectively. The average marks obtained by all the candidates who appeared for the examination is more than 43 but not more than 49. The number of candidates who passed as a fraction of the total number of candidates who appeared for the examination could never be
 (1) $\frac{3}{5}$ (2) $\frac{4}{7}$
 (3) $\frac{18}{35}$ (4) $\frac{24}{35}$
101. A library has an average of 510 visitors on Sundays and 240 on other days. The average number of visitors per day in a month of 30 days beginning with a Sunday is -
 (1) 250 (2) 276
 (3) 280 (4) 285
102. Oranges at Rs.40 per kg are mixed with oranges at Rs.80 per kg to make a mixture of 21 kg worth Rs.60 per kg. How many kg of each variety is taken?
 (1) 8 kg, 13 kg (2) 6 kg, 15 kg
 (3) 10.5 kg, 10.5 kg (4) 12 kg, 9 kg
103. Ramesh receives the following scores in his maths test: 78, 92, 83 and 99. What score does he need in the next test in order to have an average of 90 in the five tests?
 (1) 96 (2) 97
 (3) 98 (4) 99
104. If the average marks of three batches of 55, 60 and 45 students respectively is 50, 55, 60, then the average marks of all the students is -
 (1) 53.33 (2) 54.68
 (3) 55 (4) None of these
105. Rahul purchases 20 kg of wheat at the rate of Rs.9 per kg and 10 kg of wheat at the rate of Rs.12 per kg. He mixed the both types and sold the mixture. Approximately at what price per kg should he sell the mixture to make 20% profit?
 (1) Rs.10 (2) Rs.12
 (3) Rs.9 (4) Rs.13
106. The average age of a class of 60 students is 12 years. If the number of girls is increased by $\frac{3}{4}$ th and the number of boys is decreased by half, the total number of students does not change; nor does the average age of either the boys or the girls change. But the combined average age changes to 13.5 years. The average age of the boys is
 (1) 8 years (2) 10 years
 (3) 12 years (4) Cannot be determined
107. A pupil's marks were wrongly entered as 83 instead of 63. Due to that the average marks for the class got increased by half ($\frac{1}{2}$). The number of pupils in the class is -
 (1) 10 (2) 20
 (3) 40 (4) 73
108. A vessel of 90 litres is completely filled with milk and water. 60% of milk and 40% of water is taken out of the vessel. It is found that the vessel is vacated by 50%. Find the initial quantity of milk.
 (1) 45 litres (2) 50 litres
 (3) 60 litres (4) 30 litres
109. The average age of the students in an institute was 15 years. Then 10 students of average age 12 years joined the institute and the average age of the students of this institute became 14.8 years. What was the initial number of students in that institute?
 (1) 140 (2) 120
 (3) 150 (4) None of these
110. A container contains 64-liter milk. 16 liter are taken out and replaced by water and this is repeated 2 times more. Find ratio of water and milk now.
 (1) 27 : 37 (2) 9 : 7
 (3) 37 : 27 (4) 16 : 48
111. A vessel contains a mixture of two liquids A and B in the ratio 7 : 5. When 9 litres of mixture are drawn off and the vessel is filled with liquid B, the ratio of A and B becomes 7 : 9. Quantity of liquid A (in litres) contained in the vessel initially was
 (1) 10 (2) 20
 (3) 21 (4) 25
112. The monthly salaries of two persons A and B are in the ratio of 3 : 5 respectively. If both of them received an increment of Rs. 250, then the ratio becomes 2 : 3. What were their respective salaries before the increment?
 (1) Rs. 850 & Rs. 1,275 (2) Rs. 700 & Rs. 1,050
 (3) Rs. 750 & Rs. 1,250 (4) Rs. 650 & Rs. 975
113. A container of certain quantity was full of milk. 8 liter were drawn off and replaced by water. This process was repeated 3 times more. Ratio of water and milk became 65:16. Find capacity of container.
 (1) 20 (2) 24
 (3) 32 (4) 40
114. Ramesh analysed the monthly salary figures of five vice presidents of his company. All the salary figures are integers. The mean and the median salary figures are Rs 5 lakh, and the only mode is Rs 8 lakh. Which of the options

- below is the sum (in Rs lakh) of the highest and the lowest salaries?
- (1) 9 (2) 10
(3) 11 (4) 12
115. 8 liters are drawn from a cask full of wine and is then filled with water. This operation is performed three more times. The ratio of the quantity of wine now left in cask to that of water is 16 : 65. How much wine did the cask hold originally?
- (1) 18 litres (2) 24 litres
(3) 32 litres (4) 42 litres
116. Ramu appears in six different papers in his semester examination, where the maximum marks were 50 for each paper. His marks in these papers are in the proportion 8 : 9 : 10 : 13 : 14 : 15. Considering his aggregate in all the papers together, he fails to obtain 50% of the total marks. What is the minimum possible additional marks Ramu should get to obtain 50% of the total marks, given that he got integral marks in each paper?
- (1) 81 (2) 57
(3) 12 (4) 18
117. In a container, Milk and Water are in ratio 7 : 5. When 9 litres of mixture were removed and replaced by water, ratio became 7 : 9. Find initial quantity of milk and mixture. A vessel is filled with liquid, 3 parts of which are water and 5 parts syrup. How much of the mixture must be drawn off and replaced with water so that the mixture may be half water and half syrup?
- (1) $\frac{1}{3}$ (2) $\frac{1}{4}$
(3) $\frac{1}{5}$ (4) $\frac{1}{7}$
118. The average age of 7 consecutive natural numbers is K, then what will be the average of next 7 consecutive natural numbers?
- (1) K (2) K + 7
(3) K + 1 (4) K + 49
119. If average of the age of Husband and wife 6 years ago was 21 and the average age of and the average age of husband, wife and their son 3 year hence is 22 years, then what is the present age of son?
- (1) 4 years (2) 5 years
(3) 6 years (4) 3 years